

ELEC 221 Lecture 04

The convolution integral; the impulse response and system properties

Tuesday 22 September 2026

Announcements

Is $y(t) = x(4t+1)$

$x(t) \rightarrow \boxed{S} \rightarrow x(4t+1)$

- Tutorial assignment 1 due Friday 23:59
- Quiz 2 today

Time invariant? $x(t-t_0) \rightarrow y(t-t_0)$

$x'(t) = x(t-t_0) \rightarrow y'(t) = x'(4t+1) = x(4t+1-t_0)$

$y(t-t_0) = x(4(t-t_0)+1)$

not the same!

The convolution sum

We expressed signals as weighted sums of impulses

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

coeff. *function (signal)*

If we know what an LTI system does to a unit impulse (i.e., the impulse response $h[n]$), we know what it does to any other signal:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

This is the **convolution sum**. Today we will see the **convolution integral** in continuous time.

Today

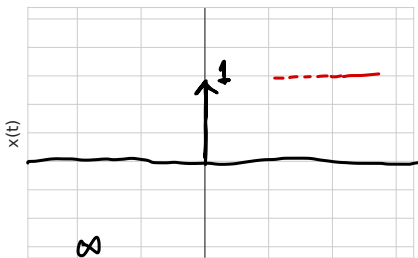
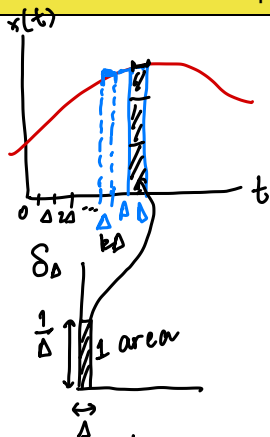
Learning outcomes:

- Define the CT unit impulse and step functions
- Define the convolution integral and use it to compute the output of a system
- Use the impulse response to determine whether a system is stable, causal, memoryless, or invertible

If we have time:

- Use the convolution sum/integral to show that complex exponentials are eigenfunctions of LTI systems

The CT unit impulse



$$\int_{-\infty}^{\infty} \delta(\tau) d\tau = 1$$

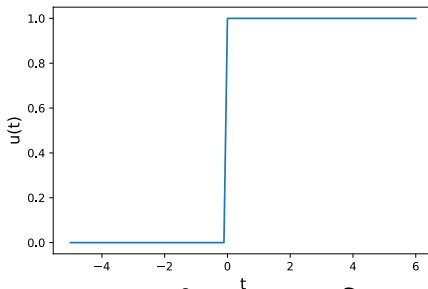
CT unit impulse

$$x[k] \delta[n-k]$$

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta) \cdot \delta_{\Delta}(t-k\Delta) \cdot \Delta$$

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau$$

The CT unit step



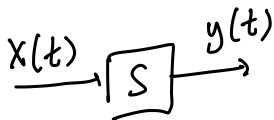
$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

Just like in DT, the unit impulse and step are related:

$$\delta(t) = \frac{du(t)}{dt}$$

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

The convolution integral



$$\delta(t) \rightarrow h(t)$$

The CT analogue of convolution sum is the **convolution integral**.

$$y(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau \rightarrow y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

where $h(t)$ is the **CT impulse response**.

DT: $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$

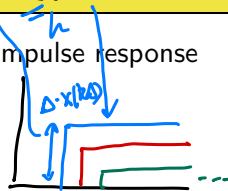
It has the same properties (commutative, associative, distributive).

Example: convolution

$$\hat{x}(t) = \sum_{k=-\infty}^{\infty} x(k\Delta) \delta_{\Delta}(t-k\Delta) \cdot \Delta$$

(Oppenheim Ex. 2.6 Var.) Consider system with impulse response

$$h(t) = u(t-1)$$

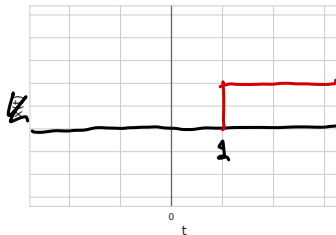


What is the output of the system for the input signal

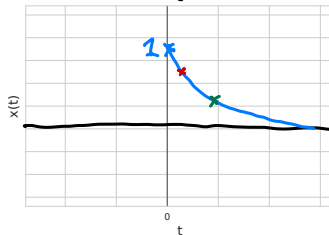
$$x(t) = e^{-2t} u(t)$$

↓ sum up
(qualitative)

$h(t)$



$x(t)$



Example: convolution

$$x(t) = e^{-2t} u(t) \quad h(t) = u(t-1)$$

$$\Rightarrow y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau$$

Non-zero: $\tau \geq 1$

$t > \tau$

otherwise
vanishes

$$= \int_{-\infty}^{\infty} e^{-2(t-\tau)} u(t-\tau) u(\tau-1) d\tau$$

$$= \int_1^{\infty} e^{-2(t-\tau)} u(t-\tau) d\tau \quad (\tau > 1)$$

$$= \int_1^t e^{-2(t-\tau)} d\tau \quad (\tau < t)$$

$$= e^{-2t} \int_1^t e^{2\tau} d\tau$$

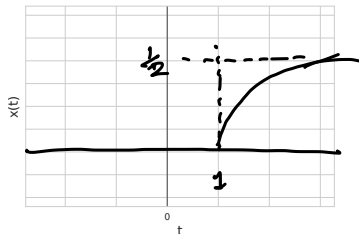
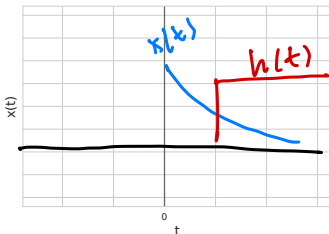
$$= e^{-2t} \left(\frac{1}{2} e^{2\tau} \Big|_1^t \right) = e^{-2t} \left(\frac{1}{2} e^{2t} - \frac{1}{2} e^2 \right) = \frac{1}{2} (1 - e^{-2(t-1)})$$

Example: convolution

$$y(t) = \frac{1}{2} (1 - e^{-2(t-1)}) \Rightarrow \text{bounds: } \begin{cases} \tau \geq 1 \\ t > \tau \end{cases}$$
$$= \frac{1}{2} (1 - e^{-2(t-1)}) u(t-1) \quad t > 1$$

←

↓



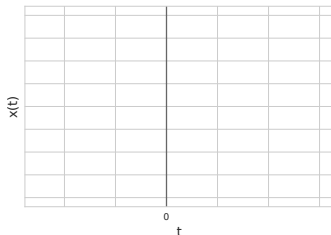
Exercise: convolution

(Oppenheim 2.8) Consider system with impulse response

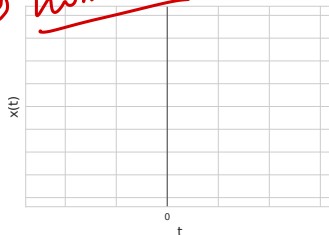
$$h(t) = \delta(t+2) + 2\delta(t+1)$$

What is the output of the system for the input signal

$$x(t) = \begin{cases} t+1 & 0 \leq t \leq 1 \\ 2-t & 1 < t \leq 2 \\ 0 & \text{elsewhere.} \end{cases}$$



⇒ homework.



Impulse response and analysis of LTI systems

To reiterate: the convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

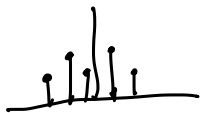
and convolution integral

$$x(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

show that as long as we know how a system responds to a unit impulse, we can determine its response to any other signal.

Knowledge of the impulse response also allows us to determine key properties of systems such as causality, invertibility, and stability.

Impulse response and memory



$$\begin{aligned} y[n] &= \sum_{k=-\infty}^{\infty} x[k] h[n-k] \\ &= \sum_{k=-\infty}^{\infty} x[n-k] h[k] \end{aligned}$$

A system is memoryless if the output depends only on the input at the same time. This implies $h[n] = 0$ for $n \neq 0$, meaning

$$h[n] = K \cdot \delta[n]$$

In CT, $h(t) = K\delta(t)$.

Impulse response and invertibility

If a system is invertible, it has an inverse system.

Suppose impulse response of a system is $h(t)$. Then

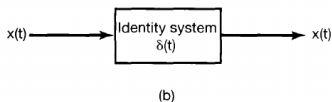
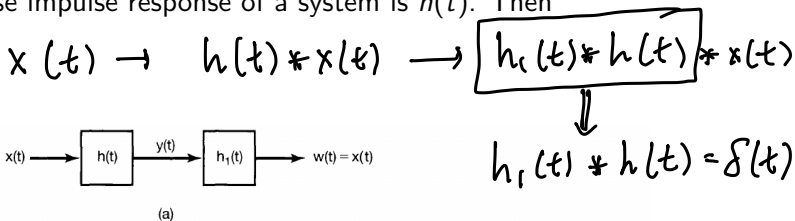


Figure 2.26 Concept of an inverse system for continuous-time LTI systems. The system with impulse response $h_1(t)$ is the inverse of the system with impulse response $h(t)$ if $h(t) * h_1(t) = \delta(t)$.

(Analogous for DT case. We will see this later in the course.)

For next time

Content: *★ Start w/ analysis of causality and stability.*

- (Complex) exponential signals
- The continuous-time Fourier series representation
- CT Fourier series representation and properties

Action items:

- ① Tutorial Assignment 1 due Friday 23:59

Recommended reading:

- From today's class: Oppenheim 1.4, 2.2-2.3
- Practice problems: 2.8-2.12, 2.14-16, 2.22, 2.28, 2.29
- For next class: Oppenheim 1.3, 3.0-3.5